Supplementary material: Sterile-Neutrino Dark Matter and Big-Bang Nucleosynthesis Constraints

HEP-Genesis-Agent session record
Project: hep-testing-research

2026-10-15

Supplementary material, generated mechanically from the session record: it lists exactly what ran, in order, without interpretation.

A Implementation details

This appendix is generated mechanically from the session record; it lists exactly what ran, in order, without interpretation.

A.1 Session

- Source: cli client; conversation `f281522f`
- Started: 2026-10-15 20:51:32; report generated: 2026-10-15 20:56:32
- Project: hep-testing-research (`/Users/nramachandra/hep-genesis-testing/projects/hep-testing-research`)
- Compute execution site(s) selected in the session: local
- 1 user turn(s), 57 tool call(s) (0 returned errors), 0 figure(s), 8 data file(s)

A.2 Models

Stage	Model	Endpoint
quick	`gpt-5.6-luna@openai`	`https://mag.genesis.americansciencecloud.org/v1`
report	`gpt-5.6-luna@openai`	`https://mag.genesis.americansciencecloud.org/v1`
coder	`gpt-5.6-luna@openai`	`https://mag.genesis.americansciencecloud.org/v1`

A.3 MCP servers

Server	Transport	Location
hep-research-aid	stdio	`python -m hep_genesis.domain.mcp_server`

Server	Transport	Location
alcf-tools	stdio	`python -m hep_genesis.iri.alcf.mcp_server`
nersc-tools	stdio	`python -m hep_genesis.iri.nersc.mcp_server`

A.4 Tool-call log

Chronological; arguments and results are truncated for space (the full record is in the project's `provenance/` folder).

#	Tool	Arguments	Outcome
1	`hep-research-aid list_agent_files`	`{}`	`[]`
2	`hep-research-aid list_input_files`	`{}`	`{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }`
3	`hep-research-aid list_datasets`	`{}`	`{'count': 0, 'datasets': []}`
4	`hep-research-aid list_files`	`{}`	`Files in .: - downloads - logs - projects - reports - runs - scripts - venvs - workspace`
5	`hep-research-aid get_dispatch`	`{}`	`Dispatch: local execution.`
6	`local search_arxiv_abstracts`	`{"query": "sterile neutrino dark matter BBN constraints 2024", "max_results": 10}`	`[1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of...`
7	`local search_arxiv_abstracts`	`{"query": "keV sterile neutrino dark matter primordial nucleosynthesis mixing", "max_results": 10}`	`[1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of...`

#	Tool	Arguments	Outcome
8	local search_arxiv_ abstracts	{"query": "sterile neutrino dark matter cosmology BBN 2025", "max_results": 10}	[1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of...
9	hep-research-aid search_arxiv	{"query": "sterile neutrino BBN heavy neutrino decay", "max_results": 10, "sort_ by": "submittedDate"}	Found 10 papers matching query: 'sterile neutrino BBN heavy neutrino decay' (source: arXiv API) Paper 1: arXiv ID: 2610.21584v1 Title: Geometric Queries on Tessellation-Free Displaced Surfaces Author...
10	hep-research-aid search_arxiv	{"query": "sterile neutrino dark matter X-ray Lyman alpha 2024", "max_ results": 10, "sort_by": "submittedDate"}	Found 10 papers matching query: 'sterile neutrino dark matter X-ray Lyman alpha 2024' (source: arXiv API) Paper 1: arXiv ID: 2610.21589v1 Title: Probing fundamental physics with ALMA2040 mm-wavelengt...
11	hep-research-aid search_arxiv	{"query": "sterile neutrino dark matter primordial nucleosynthesis", "max_ results": 10, "sort_by": "submittedDate"}	Found 10 papers matching query: 'sterile neutrino dark matter primordial nucleosynthesis' (source: arXiv API) Paper 1: arXiv ID: 2610.21589v1 Title: Probing fundamental physics with ALMA2040 mm-wavel...
12	hep-research-aid search_arxiv	{"query": "right handed neutrino dark matter BBN 2025", "max_results": 10, "sort_by": "submittedDate"}	Found 10 papers matching query: 'right handed neutrino dark matter BBN 2025' (source: arXiv API) Paper 1: arXiv ID: 2610.21613v1 Title: To Think or Not to Think: Pre-Decisional Reasoning Budgets for ...

#	Tool	Arguments	Outcome
13	local get_arxiv_ metadata	{"arxiv_id": "2410.07343"}	arXiv ID: 2410.07343 Title: BBN Constraint on Heavy Neutrino Production and Decay Authors: Yu-Ming Chen, Yue Zhang Year: 2024 Abstract: We explore the big-bang nucleosynthesis (BBN) constraint on heav...
14	local search_arxiv_ abstracts	{"query": "\"sterile neutrino dark matter\" 2024 X-ray", "max_results": 10}	[1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of...
15	local search_arxiv_ abstracts	{"query": "\"BBN Constraint on Heavy Neutrino Production and Decay\"", "max_results": 5}	[2410.07343v2] BBN Constraint on Heavy Neutrino Production and Decay Abstract: We explore the big-bang nucleosynthesis (BBN) constraint on heavy neutrino that is a mixture of gauge singlet fermion and...
16	local search_arxiv_ abstracts	{"query": "sterile neutrino warm dark matter constraints 2024 2025", "max_results": 10}	[1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of...
17	local search_arxiv_ abstracts	{"query": "sterile neutrino dark matter resonant production mixed warm cold X-ray Lyman alpha 2306.16532", "max_results": 5}	[1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of...

#	Tool	Arguments	Outcome
18	local search_arxiv_ abstracts	{"query": "sterile neutrino dark matter neutrino self interactions 2112.00758", "max_results": 5}	[0211001v1] Very Long Baseline Neutrino Oscillation Experiment for Precise Determination of Oscillation Parameters and Search for nu_mu -> nu_e Appearance and CP Violation Abstract: The possibility ...
19	local search_arxiv_ abstracts	{"query": "freeze-in freeze-out sterile neutrino dark matter 2204.08795", "max_results": 5}	[2204.08795v3] Freeze-in and freeze-out of sterile neutrino dark matter Abstract: A sterile neutrino with a keV-scale mass is a compelling dark matter candidate. We propose a new production mechanism ...
20	local get_arxiv_ metadata	{"arxiv_id": "2410.07343"}	arXiv ID: 2410.07343 Title: BBN Constraint on Heavy Neutrino Production and Decay Authors: Yu-Ming Chen, Yue Zhang Year: 2024 Abstract: We explore the big-bang nucleosynthesis (BBN) constraint on heav...
21	local get_arxiv_ metadata	{"arxiv_id": "1702.08430"}	arXiv ID: 1702.08430 Title: keV sterile neutrino Dark Matter Authors: Alexander Merle Year: 2017 Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short ...
22	local download_arxiv_ full_text	{"arxiv_id": "2306.16532", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ hep-testing-research/runs/ 261015_205132"}	Downloaded: / Users/nramachandra/ hep-genesis-testing/ projects/ hep-testing-research/ runs/261015_205132/ 2306.16532.txt

#	Tool	Arguments	Outcome
23	local download_arxiv_full_text	{"arxiv_id": "2112.00758", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2112.00758.txt
24	local download_arxiv_full_text	{"arxiv_id": "2204.08795", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2204.08795.txt
25	local download_arxiv_full_text	{"arxiv_id": "2410.07343", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2410.07343.txt
26	local download_arxiv_full_text	{"arxiv_id": "1702.08430", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/1702.08430.txt
27	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2306.16532.txt", "start": 0, "length": 14000}	[chars 0-14000 of 47413] (more available, next start=14000) Title: Exploring resonantly produced mixed sterile neutrino dark matter models Authors: Emma L. Horner, Francisco Mungia Wulftange, Isabell...
28	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2112.00758.txt", "start": 0, "length": 12000}	[chars 0-12000 of 59548] (more available, next start=12000) Title: Sterile neutrino dark matter production in presence of non-standard neutrino self-interactions: an EFT approach Authors: Cristina Be...

#	Tool	Arguments	Outcome
29	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2204.08795.txt", "start": 0, "length": 12000}	[chars 0-12000 of 87661] (more available, next start=12000) Title: Freeze-in and freeze-out of sterile neutrino dark matter Authors: Rupert Coy, Michael A. Schmidt Year: 2022 arXiv ID: 2204.08795 ==...
30	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2410.07343.txt", "start": 0, "length": 12000}	[chars 0-12000 of 88681] (more available, next start=12000) Title: BBN Constraint on Heavy Neutrino Production and Decay Authors: Yu-Ming Chen, Yue Zhang Year: 2024 arXiv ID: 2410.07343 ==========...
31	board post_finding	{"claim": "The 2024 Chen-Zhang minimal HNL analysis treats (m4, |Ua4| ^2) as the only parameters and finds BBN exclusion only in an intermediate-mixing window: large mixing decays before BBN, while tiny mixing produces to...	Recorded finding f-001.
32	board post_finding	{"claim": "Minimal Dodelson-Widrow sterile-neutrino production that supplies all dark matter is excluded by X-ray constraints in the cited self-interaction study, while nonstandard active-neutrino self-interactions can w...	Recorded finding f-002.
33	board post_finding	{"claim": "Resonant Shi-Fuller production requires an initial neutrino lepton number many orders of magnitude above the baryon-to-photon ratio but below detectable CMB/BBN limits in the 7.1-keV mixed cold-plus-warm study...	Recorded finding f-003.

#	Tool	Arguments	Outcome
34	board post_finding	{"claim": "The scalar-mediated freeze-in/ freeze-out benchmark produces keV sterile dark matter through a two-step process: Higgs-portal freeze-in of a complex scalar sector followed by sterile-neutrino production from sc...	Recorded finding f-004.
35	local read_arxiv_chunk	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ hep-testing-research/ runs/261015_205132/ 2306.16532.txt", "start": 30000, "length": 12000}	[chars 30000-42000 of 47413] (more available, next start=42000) er spectra than resonant, so we can see that the number of subhalos decreases more rapidly with increasing mixing angle than with incre...
36	local read_arxiv_chunk	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ hep-testing-research/ runs/261015_205132/ 2410.07343.txt", "start": 50000, "length": 12000}	[chars 50000-62000 of 88681] (more available, next start=62000) nge region is for temporary matter domination. In the blue, green and red regions, the heavy neutrino have different dominant decay mod...
37	local read_arxiv_chunk	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ hep-testing-research/ runs/261015_205132/ 1702.08430.txt", "start": 0, "length": 10000}	[chars 0-10000 of 21223] (more available, next start=10000) Title: keV sterile neutrino Dark Matter Authors: Alexander Merle Year: 2017 arXiv ID: 1702.08430 ==========...==========...
38	hep-research-aid arxiv_agent	{"query": "Survey arXiv 2024-2026 papers directly analyzing sterile neutrino dark matter and Big Bang nucleosynthesis BBN, including keV resonant production, X-ray and Lyman-alpha constraints, and MeV-GeV heavy neutral l...	ARXIV SURVEY: STERILE-NEUTRINO DARK MATTER, BBN, AND HNL CONSTRAINTS Search coverage: arXiv submissions through 2026-10-14, based on the returned arXiv abstracts. Full-text extraction was attempted bu...

#	Tool	Arguments	Outcome
39	board post_finding	{"claim": "In the 7.1-keV mixed cold-plus-warm resonant-production calculation, models retaining at least 60 Milky-Way satellite subhalos include nearly 100% sterile DM with astrophysical mixing sin^2(2θastro) about 7$\times$10...	Recorded finding f-005.
40	hep-research-aid search_arxiv	{"query": "sterile neutrino dark matter BBN 2025", "max_results": 20, "sort_by": "submittedDate"}	Found 20 papers matching query: 'sterile neutrino dark matter BBN 2025' (source: arXiv API) Paper 1: arXiv ID: 2610.21589v1 Title: Probing fundamental physics with ALMA2040 mm-wavelength observations...
41	hep-research-aid search_arxiv	{"query": "heavy neutral lepton BBN sterile neutrino 2024 2025", "max_results": 20, "sort_by": "submittedDate"}	Found 20 papers matching query: 'heavy neutral lepton BBN sterile neutrino 2024 2025' (source: arXiv API) Paper 1: arXiv ID: 2610.21609v1 Title: Fast, parallel, three-dimensional rearrangement of ato...
42	local search_arxiv_abstracts	{"query": "keV sterile neutrino dark matter BBN resonant production X-ray Lyman alpha", "max_results": 20}	[1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of...
43	local get_arxiv_metadata	{"arxiv_id": "2410.07343"}	arXiv ID: 2410.07343 Title: BBN Constraint on Heavy Neutrino Production and Decay Authors: Yu-Ming Chen, Yue Zhang Year: 2024 Abstract: We explore the big-bang nucleosynthesis (BBN) constraint on heav...

#	Tool	Arguments	Outcome
44	local get_arxiv_ metadata	{"arxiv_id": "2507.12270"}	arXiv ID: 2507.12270 Title: No Hiding in the Dark: Cosmological Bounds on Heavy Neutral Leptons with Dark Decay Channels Authors: P. S. Bhupal Dev, Quan-feng Wu, Xun-Jie Xu Year: 2025 Abstract: Heavy ...
45	local get_arxiv_ metadata	{"arxiv_id": "2410.06970"}	arXiv ID: 2410.06970 Title: Relaxing Limits from Big Bang Nucleosynthesis on Heavy Neutral Leptons with Axion-like Particles Authors: Frank F. Deppisch, Tomás E. Gonzalo, Chayan Majumdar, Zhong Zhang ...
46	local get_arxiv_ metadata	{"arxiv_id": "2510.01907"}	arXiv ID: 2510.01907 Title: Resonant production of sterile neutrino dark matter with a refined numerical scheme Authors: Kentaro Kasai, Masahiro Kawasaki, Kai Murai Year: 2025 Abstract: The existence ...
47	local get_arxiv_ metadata	{"arxiv_id": "2402.11902"}	arXiv ID: 2402.11902 Title: Affleck-Dine leptogenesis scenario for resonant production of sterile neutrino dark matter Authors: Kentaro Kasai, Masahiro Kawasaki, Kai Murai Year: 2024 Abstract: Sterile...
48	local download_arxiv_ full_text	{"arxiv_id": "2410.07343", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ hep-testing-research/runs/ 261015_205132"}	Downloaded: / Users/nramachandra/ hep-genesis-testing/ projects/ hep-testing-research/ runs/261015_205132/ 2410.07343.txt

#	Tool	Arguments	Outcome
49	local download_arxiv_full_text	{"arxiv_id": "2507.12270", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2507.12270.txt
50	local download_arxiv_full_text	{"arxiv_id": "2410.06970", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2410.06970.txt
51	local download_arxiv_full_text	{"arxiv_id": "2510.01907", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2510.01907.txt
52	board post_finding	{"claim": "The directly relevant 2024 minimal-model analysis arXiv:2410.07343 (Chen and Zhang, BBN Constraint on Heavy Neutrino Production and Decay) parameterizes heavy-neutrino cosmology solely by mass m4 and flavor-sp...	Recorded finding f-006.
53	board post_finding	{"claim": "The 2025 paper arXiv:2507.12270 states that for thermally equilibrated sub-GeV HNLs, successful BBN requires lifetime tau_N less than approximately 0.02 s; significant dark decay modes do not generically evade...	Recorded finding f-007.
54	board post_finding	{"claim": "The 2024 paper arXiv:2410.06970 finds a nonminimal HNL-plus-axion-like-particle decay scenario with a new cosmologically consistent region approximately 1 MeV-1 GeV in HNL mass and active-sterile mixing streng...	Recorded finding f-008.

#	Tool	Arguments	Outcome
55	board post_finding	{"claim": "The 2025 resonant-production analysis arXiv:2510.01907 reports keV sterile-neutrino dark-matter parameter regions consistent with X-ray and Lyman-alpha constraints for lepton-to-entropy ratio greater than or a...	Recorded finding f-009.
56	board post_finding	{"claim": "The 2024 Affleck-Dine resonant-production paper arXiv:2402.11902 studies Shi-Fuller keV sterile-neutrino dark matter generated by a large lepton asymmetry from an Affleck-Dine scenario with an L-ball, and revi...	Recorded finding f-010.
57	board post_dead_end	{"what": "Attempted autonomous recent-literature survey and full-text extraction for selected 2024-2026 sterile-DM/BBN/HNL papers via arxiv_agent.", "why": "The autonomous agent reported RuntimeError: 'Search' object has...	Recorded dead end x-01.

A.5 Files produced

- 2306.16532.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2306.16532.txt
- 2112.00758.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2112.00758.txt
- 2204.08795.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2204.08795.txt
- 2410.07343.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2410.07343.txt
- 1702.08430.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/1702.08430.txt
- 2507.12270.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2507.12270.txt
- 2410.06970.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2410.06970.txt

- `2510.01907.txt` (text, download_arxiv_full_text): `/Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2510.01907.txt`

B Research board

The shared board of the research runs in this session, generated mechanically: what the team set out to investigate, what it found (with the tool calls that are the evidence), what failed, and how the critic judged each wave.

B.1 Directions

Id	Status	Goal	Outcome
`d-01`	closed	Survey 2024–2026 arXiv literature directly addressing sterile-neutrino dark matter, BBN constraints, and the surviving mass–mixing parameter regions.	Survey established several relevant 2024–2025 papers. The directly relevant minimal HNL-BBN analysis arXiv:2410.07343 uses (m4, |U_a4|^2) separately for a=e, μ, τ and includes temporary matter domination plus meson, photon, and neutrino injection; full text saved at /Users/nramachandra/hep-genesis-testing/projects/hep-testing-research/runs/261015_205132/2410.07343.txt. arXiv:2507.12270 reports tau
`d-02`	closed	Build a cross-checked phenomenological map of viable sterile-neutrino parameter space by comparing recent BBN analyses with the established X-ray, relic-density, and structure-formation constraints.	Established a cross-checked phenomenological map. For keV sterile DM, minimal Dodelson–Widrow production supplying all DM is excluded by X-ray limits in arXiv:2112.00758 and is also disfavored by structure formation; nonstandard neutrino self-interactions can shift/widen the abundance-compatible mixing band, while a subdominant sterile fraction relaxes X-ray constraints. Resonant Shi–Fuller produc
`d-03`	open	Reproduce the electron-, muon-, and tau-mixing BBN contours of arXiv:2410.07343, including temporary matter domination and decay-injection effects.	
`d-04`	open	Construct a conditional keV sterile-neutrino scan combining relic abundance, radiative-decay X-rays, momentum distribution, free streaming, and a specified structure-formation baseline.	

Id	Status	Goal	Outcome
d-05	open	Compare refined resonant-production histories across the 7.1-keV mixed-component benchmark and the m_s approximately above 20 keV, lepton-asymmetry-driven region.	
d-06	open	Extend the HNL analysis to ALP-cascade and dark-decay scenarios with explicit lifetime, abundance, decay-temperature, and radiation-density calculations.	
d-07	open	Quantify the abundance–mixing relation in non-standard neutrino self-interaction and scalar freeze-in/freeze-out models.	

B.2 Findings and evidence

Id	Confidence	Claim	Evidence (tool calls, figures)
f-001	high, **contested**	The 2024 Chen–Zhang minimal HNL analysis treats (m4, |Ua4|^2) as the only parameters and finds BBN exclusion only in an intermediate-mixing window: large mixing decays before BBN, while tiny mixing produces too little freeze-in abundance to affect light elements; it includes temporary matter domination and charged-meson, photon, and neutrino injection, with a separate CMB constraint for very long-lived states.	tc-261015_2-w1-d-02-21, tc-261015_2-w1-d-02-31 Fig. 1
f-002	high, **contested**	Minimal Dodelson–Widrow sterile-neutrino production that supplies all dark matter is excluded by X-ray constraints in the cited self-interaction study, while nonstandard active-neutrino self-interactions can widen or shift the correct-abundance region toward smaller or larger mixing; allowing a subdominant 'cocktail' fraction further relaxes X-ray limits, and the study reports no structure-formation problem from the extra interactions in its benchmark framework.	tc-261015_2-w1-d-02-29
f-003	high, **contested**	Resonant Shi–Fuller production requires an initial neutrino lepton number many orders of magnitude above the baryon-to-photon ratio but below detectable CMB/BBN limits in the 7.1-keV mixed cold-plus-warm study; the paper computes the distribution by Boltzmann evolution including the density potential proportional to lepton asymmetry and assesses viability through satellite-host subhalo formation rather than a full BBN abundance fit.	tc-261015_2-w1-d-02-28

Id	Confidence	Claim	Evidence (tool calls, figures)
f-004	high, **contested**	The scalar-mediated freeze-in/freeze-out benchmark produces keV sterile dark matter through a two-step process: Higgs-portal freeze-in of a complex scalar sector followed by sterile-neutrino production from scalar decays and annihilations; the hidden sector can thermalize at a temperature below the SM temperature, changing the momentum distribution and free-streaming relative to oscillation production.	tc-261015_2-w1-d-02-30 Fig. 1
f-005	high, **contested**	In the 7.1-keV mixed cold-plus-warm resonant-production calculation, models retaining at least 60 Milky-Way satellite subhalos include nearly 100% sterile DM with astrophysical mixing sin^2(2θastro) about 7×10^-10; viable models have sin^2(2θastro) less than roughly 10^-9. Here f is the sterile fraction and sin^2(2θastro)=f sin^2(2θtot), so X-ray-inferred mixing differs from microscopic mixing when f<1.	tc-261015_2-w1-d-02-36 Fig. 11, 12
f-006	high, **contested**	The directly relevant 2024 minimal-model analysis arXiv:2410.07343 (Chen and Zhang, BBN Constraint on Heavy Neutrino Production and Decay) parameterizes heavy-neutrino cosmology solely by mass m4 and flavor-specific mixing |U_a4|^2 for a=e, μ, τ; it computes the full BBN-excluded space including temporary matter domination and injection of charged mesons, photons, and light neutrinos, and presents BBN as complementary to terrestrial HNL searches.	tc-261015_2-w1-d-01-43
f-007	high, **contested**	The 2025 paper arXiv:2507.12270 states that for thermally equilibrated sub-GeV HNLs, successful BBN requires lifetime tau_N less than approximately 0.02 s; significant dark decay modes do not generically evade the bound and instead strengthen cosmological constraints through extra radiation affecting primordial helium and ΔN_eff.	tc-261015_2-w1-d-01-44, tc-261015_2-w1-d-01-49

Id	Confidence	Claim	Evidence (tool calls, figures)
f-008	high, **contested**	The 2024 paper arXiv:2410.06970 finds a nonminimal HNL-plus-axion-like-particle decay scenario with a new cosmologically consistent region approximately 1 MeV–1 GeV in HNL mass and active-sterile mixing strength approximately 10^-9–10^-6; the HNL decays primarily to ALP plus neutrino early enough to relax ordinary BBN limits, while later ALP decay remains subject to cosmological and astrophysical constraints.	tc-261015_2-w1-d-01-45, tc-261015_2-w1-d-01-50
f-009	high, **contested**	The 2025 resonant-production analysis arXiv:2510.01907 reports keV sterile-neutrino dark-matter parameter regions consistent with X-ray and Lyman-alpha constraints for lepton-to-entropy ratio greater than or approximately order 10^-3 and sterile-neutrino mass m_nus greater than approximately 20 keV; production during lepton-number injection can shorten free-streaming length and reduce the mixing needed for the observed abundance.	tc-261015_2-w1-d-01-46, tc-261015_2-w1-d-01-51
f-010	high	The 2024 Affleck-Dine resonant-production paper arXiv:2402.11902 studies Shi-Fuller keV sterile-neutrino dark matter generated by a large lepton asymmetry from an Affleck-Dine scenario with an L-ball, and revisits X-ray and Lyman-alpha constraints; the abstract does not state a numerical surviving mass-mixing interval.	tc-261015_2-w1-d-01-47

B.3 Dead ends

- x-01 (d-01): Attempted autonomous recent-literature survey and full-text extraction for selected 2024–2026 sterile-DM/BBN/HNL papers via arxiv_agent. — The autonomous agent reported RuntimeError: 'Search' object has no attribute 'results' for all selected downloads, so it produced no valid files. Separate direct full-text downloads succeeded for four papers, but the tool budget ended before reading chunks or extracting figures and captions.

B.4 Critic verdicts

- c-01 (wave 0, plan): **revise**; high: The direction risks conflating the 2024 BBN analysis of MeV–GeV heavy neutral leptons with keV sterile-neutrino dark matter. It must explicitly classify each paper by mass regime, lifetime, abundance, and production mechanism before combining results.; medium: The requested quantitative surviving regions may not be recoverable from published plots or captions with the available tools. Do not report interpolated numerical boundaries as published results; label plot-read values as approximate and preserve the source assumptions.; medium: The search plan needs an auditable coverage record: exact queries, arXiv date cutoff, inclusion and exclusion criteria,

and checks for versions or papers whose titles do not mention BBN. This is important if the conclusion is that no 2025–2026 keV-DM-plus-BBN paper exists.; medium: The 19-call budget is not operationalized. The plan should prioritize a small set of primary papers and reserve calls for full-text verification and citation extraction rather than attempting exhaustive downloads, figure extraction, and duplicate searches.; high: A comparison matrix cannot generally establish joint viability by marking independent constraints as passed or failed. X-ray, relic-density, lepton-asymmetry, momentum-distribution, and structure limits can share correlated production assumptions. The synthesis must distinguish published combined sc; medium: The direction should not imply a universal mass–mixing band for resonant, freeze-in, freeze-out, or self-interaction scenarios. These mechanisms can decouple abundance from mixing, so results should be reported as conditional benchmark points or regions with all extra parameters stated.

- `c-02` (wave 1, wave): **continue**; contested `f-001`, `f-002`, `f-003`, `f-004`, `f-005`, `f-006`, `f-007`, `f-008`, `f-009`; high: The evidence ledger is not sufficient for the requested quantitative answer: most quoted excerpts are metadata or truncated abstracts, while the claims depend on numerical regions and plotted contours. Full-text passages, figure captions, and source assumptions must be quoted before treating these f; high: The synthesis mixes keV dark-matter production with MeV–GeV late-decaying HNL BBN constraints and with nonminimal production models. The final map must separate mass regime, abundance fraction, production mechanism, lifetime, and whether a point is a published joint constraint or an assembled condit

- `c-03` (wave 1, report): **revise**; high: unsupported: Figure 1 shows the published BBN-excluded regions in the heavy-neutrino mass–mixing plane, including the temporary-matter-domination region, decay-mode-dependent limits, thermalization boundaries, and representative lifetime contours. — The board and ledger support Figure 1 and the incl; high: unsupported: arXiv:2410.01907? No such paper is in the record — The board identifies arXiv:2510.01907 as the relevant resonant-production paper but does not establish the asserted absence of arXiv:2410.01907. This should be presented as a search-record observation with explicit evidence or omitted.; high: unsupported: Open directions 1 and 2 are labeled d-02 and d-01 — The board marks d-01 and d-02 as closed directions, although their results identify residual extraction and reproduction work. The report should frame these as follow-up tasks or new directions rather than reopening the closed directio

B.5 Open directions

1. `d-03` Reproduce the electron-, muon-, and tau-mixing BBN contours of arXiv:2410.07343, including temporary matter domination and decay-injection effects. — The current record establishes the qualitative exclusion structure but does not provide the numerical flavor-specific mass–mixing boundaries.

2. `d-04` Construct a conditional keV sterile-neutrino scan combining relic abundance, radiative-decay X-rays, momentum distribution, free streaming, and a specified structure-formation baseline. — The surviving keV regions depend on correlated production and structure assumptions and cannot yet be represented by one quantitative joint map.

3. `d-05` Compare refined resonant-production histories across the 7.1-keV mixed-component benchmark and the m_s approximately above 20 keV, lepton-asymmetry-driven region. — Both regions are conditionally viable, but their different lepton-asymmetry histories and structure criteria prevent a direct comparison.

4. `d-06` Extend the HNL analysis to ALP-cascade and dark-decay scenarios with explicit lifetime, abundance, decay-temperature, and radiation-density calculations. — The 2024–2025 literature shows that additional decay channels can substantially change BBN limits in the MeV–GeV regime.

5. `d-07` Quantify the abundance–mixing relation in non-standard neutrino self-interaction and scalar freeze-in/freeze-out models. — These mechanisms can reopen parameter space, but their viability remains model-dependent and must be tested jointly against X-ray and structure constraints.